Lucas A. Saavedra, Eng.

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Education

2018 – 2023 **Software engineering**, Pontifical Catholic University of Argentina (UCA), Buenos Aires, Argentina.
Final work: *Comparison of Machine Learning models for the analysis of anomalous diffusion of acetylcholine neurotransmitter receptor trajectories.*

Experience

January 2021 – June 2022 **Back-End Developer.** Deloitte, Buenos Aires, Argentina.

- Threat Intelligence Back-End Developer.
- I worked developing Codex Gigas, a malware "DNA" profiling search engine discovers malware patterns and characteristics assisting individuals who are attracted in malware hunting, and ImageSimilarity, a tool that divides a group of images according to their similarity grade using the SSIM method. Also, I developed ThreatConnect apps.
- Python, Docker, MongoDB, and SQL were used for development.

January 2022 – To date **Research Assistant.** Molecular Neurobiology Laboratory, BIOMED UCA-CONICET, (Prof. Francisco J. Barrantes, head), Pontifical Catholic University of Argentina (UCA), Buenos Aires, Argentina.

- Implementation of Deep Learning models for the analysis of static and dynamic properties of the receptor for the neurotransmitter acetylcholine (nAChR) with orientation to graph theory, artificial neural networks and geometric computing.
- Analysis of distribution and density of nicotinic acetylcholine receptors (nAChRs) in the plasma membrane of CHO-K1/A5 cells using super-resolution microscopy (STED and STORM). Advanced analytical methods implemented with MATLAB, based on graph theory and computational geometry to study the topography of nanoclusters.
- Development of efficient and autonomous algorithm based on binary classifications with supervised Graph Neural Networks (GNNs) to detect and quantify the dynamic clustering of particles in membrane proteins in single-molecule microscopy (SMLM) data. The algorithm was developed with TensorFlow.

March 2023 – To date **Assistant professor.** Introduction to Data Science, Pontifical Catholic University of Argentina (UCA), Buenos Aires, Argentina.

- Definitions and concepts of Artificial Intelligence.
- Introduction to artificial neural networks.
- Scientific method.

Experience (continued)

July 2023 – December 2024

Assistant professor. Data oriented programming, Pontifical Catholic University of Argentina (UCA), Buenos Aires, Argentina.

- Object Oriented Programming (OOP).
- Technologies: NumPy , Pandas, Python, Java, SQL.
- Software design with UML.

March 2024 – June 2024

Assistant professor. Discrete mathematics, Pontifical Catholic University of Argentina (UCA), Buenos Aires, Argentina.

- Number theory.
- Graph theory.
- Demonstration methods.
- Boolean algebra.

Research Publications

Full-Length Publications in International Peer Reviewed Journals

- **L. A. Saavedra**, A. Mosqueira, and F. J. Barrantes, "A supervised graph-based deep learning algorithm to detect and quantify clustered particles," *Nanoscale*, vol. 16, no. 32, pp. 15 308–15 318, 2024, ISSN: 2040-3364. DOI: 10.1039/D4NR01944J.
- **L. A. Saavedra**, H. Buena-Maizón, and F. J. Barrantes, "Mapping the nicotinic acetylcholine receptor nanocluster topography at the cell membrane with sted and storm nanoscopies," *International Journal of Molecular Sciences*, vol. 23, no. 18, p. 22, 2022. DOI: 10.3390/ijms231810435.

Under review

- G. Muñoz-Gil, H. Bachimanchi, J. Pineda, *et al.*, "Quantitative evaluation of methods to analyze motion changes in single-particle experiments," *Nature Communications*, 2025.
- F. Reina, **L. A. Saavedra**, C. Eggeling, and F. J. Barrantes, "Concurrent diffusion of nicotinic acetylcholine receptors and fluorescent cholesterol disclosed by two-colour sub-millisecond minflux-based single-molecule tracking," *Nature Communications*, 2025. DOI: 10.21203/rs.3.rs-5619606/v1.
- **L. A. Saavedra** and F. J. Barrantes, "Temporal convolutional networks work as general feature extractors for single-particle diffusion analysis," *J. Phys. Photonics*, 2025.

Presentations at International Conferences

- **L. A. Saavedra** and F. J. Barrantes, "Machine learning empowers static and dynamics clustering characterization of transmembrane receptors," in *Building Bridges in Computational Biophysics (BBCB) v3.0*, 2024.
- F. J. Barrantes, A. Mosqueira, H. Buena-Maizón, and **L. A. Saavedra**, "The individual molecule and its tribe, the nanocluster: The case of the nicotinic acetylcholine receptor," in *11th International Weber Symposium*, Punta del Este, Uruguay, 2023.

Miscellaneous Experience

Certification

2022 **AWS Certified Cloud Practitioner**

Courses

2020
- **The Data Scientist's Toolbox**, Johns Hopkins University, Coursera
- **Neural Networks and Deep Learning**, DeepLearning.AI, Coursera

2021
- **Machine Learning**, Stanford Online, Coursera
- **Programming Foundations: Design Patterns (2013)**, Linkedin Learning
- **R Programming**, Johns Hopkins University, Coursera

2022
- **Natural Language Processing in TensorFlow**, DeepLearning.AI, Coursera
- **DeepLearning.AI Tensorflow Developer**, DeepLearning.AI, Coursera
- **Data Science**, Coderhouse

2024
- **Deep Neural Networks with PyTorch**, IBM, Coursera

2025
- **Introduction to Statistics**, Stanford Online, Coursera